

Problem 11.12

If the effective rate of discount in year k is equal to $0.01k + 0.06$ for $k = 1, 2, 3$, find the equivalent rate of simple interest over the three-year period.

Solution

The effective discount rates for the three years are

$$d_1 = 0.01(1) + 0.06 = 0.07,$$

$$d_2 = 0.01(2) + 0.06 = 0.08,$$

and

$$d_3 = 0.01(3) + 0.06 = 0.09.$$

An effective discount rate d_k in year k gives an accumulation factor of $1/(1 - d_k)$ for that year. Therefore, the accumulation factor over the three-year period is

$$a(3) = \frac{1}{(1 - d_1)(1 - d_2)(1 - d_3)}.$$

Substituting the discount rates gives

$$a(3) = \frac{1}{(0.93)(0.92)(0.91)} \approx 1.2843631357.$$

Let i be the equivalent annual simple interest rate. Under simple interest, the accumulation factor over three years is

$$1 + 3i.$$

Equating the accumulation factors,

$$1 + 3i = \frac{1}{(0.93)(0.92)(0.91)}.$$

Solving for i ,

$$i = \frac{1}{3} \left(\frac{1}{(0.93)(0.92)(0.91)} - 1 \right) \approx 0.0947877119.$$

Hence, the equivalent annual simple interest rate is

$$\boxed{i \approx 9.47877119\%}.$$